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REGRESSION RESULTS

               Generalized Linear Model Regression Results
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Dep. Variable:                    low   No. Observations:                  189
Model:                            GLM   Df Residuals:                      185
Model Family:                 Poisson   Df Model:                            3
Link Function:                    Log   Scale:                          1.0000
Method:                          IRLS   Log-Likelihood:                -123.02
Date:                Sun, 25 Oct 2026   Deviance:                       128.04
Time:                        12:53:23   Pearson chi2:                     129.
No. Iterations:                     5   Pseudo R-squ. (CS):            0.04821
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const         -1.5527      0.205     -7.562      0.000      -1.955      -1.150
smoke          0.4350      0.261      1.665      0.096      -0.077       0.947
ht             0.8091      0.411      1.966      0.049       0.003       1.616
ui             0.6340      0.314      2.022      0.043       0.019       1.249
==============================================================================

CONFUSION MATRIX
Rows = Actual (0, 1), Columns = Predicted (0, 1)
[[119  11]
[ 45  14]]
```

For the logistic regression:

What is the model's accuracy?

70.37%

What is the model's lowest precision?

Class 1 at 56%

Do all categories have at least 30 support? (confusion matrix)

Yes

Based on the pseudo-R^2 value, is this model good in this scenario?

The R^2 value in this model is .04821 which

REGRESSION RESULTS

Generalized Linear Model Regression Results

Dep. Variable:	mhlth_crudeprev	No. Observations:	2299
Model:	GLM	Df Residuals:	2287
Model Family:	Poisson	Df Model:	11
Link Function:	Log	Scale:	1.0000
Method:	IRLS	Log-Likelihood:	-5397.6
Date:	Sun, 25 Oct 2026	Deviance:	39.316
Time:	12:58:50	Pearson chi2:	39.2
No. Iterations:	4	Pseudo R-squ. (CS):	0.1925
Covariance Type:	nonrobust		

	coef	std err	z	P>|z|	[0.025	0.975]
const	1.7032	0.172	9.888	0.000	1.366	2.041
binge_crudeprev	0.0032	0.003	0.954	0.340	-0.003	0.010
checkup_crudeprev	-0.0004	0.002	-0.192	0.848	-0.004	0.004
csmoking_crudeprev	0.0043	0.004	1.100	0.271	-0.003	0.012
depression_crudeprev	0.0137	0.002	5.800	0.000	0.009	0.018
obesity_crudeprev	0.0001	0.002	0.076	0.940	-0.003	0.004
sleep_crudeprev	0.0024	0.002	1.059	0.289	-0.002	0.007
selfcare_crudeprev	-0.0608	0.022	-2.708	0.007	-0.105	-0.017
indeplive_crudeprev	0.0522	0.014	3.759	0.000	0.025	0.079
loneliness_crudeprev	0.0113	0.003	3.717	0.000	0.005	0.017
foodinsecu_crudeprev	-0.0083	0.007	-1.247	0.212	-0.021	0.005
housinsecu_crudeprev	0.0150	0.008	1.786	0.074	-0.001	0.032

VARIANCE INFLACTION FACTOR

variable	VIF
binge_crudeprev	2.528842

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  checkup_crudeprev  1.750444
 csmoking_crudeprev  6.934794
depression_crudeprev  2.216376
  obesity_crudeprev  2.573017
    sleep_crudeprev  3.209334
 selfcare_crudeprev 30.514413
indeplive_crudeprev 30.953309
loneliness_crudeprev  3.432565
foodinsecu_crudeprev 62.830787
housinsecu_crudeprev 46.839954
```

For the count-variable regression:

Your model should include 4 significant predictors. What are they?

Depression_crudeprev, indeplive_crudeprev, loneliness_crudeprev, and selfcare_crudeprev

Which of these predictors is the most significant (hint: look at the 'coef' column)

depression_crudeprev

Do any of your significant predictors have an overinflated VIF (VIF > 10)?

Yes, selfcare_crudeprev and indeplive_crudeprev

How much of the variability is captured by the model (hint: look at pseudo-r^2)

The model captures majority of the variability with the R^2 being 0.1925